

School of Natural Sciences and Mathematics

The Biology Program at UT Dallas emphasizes the unifying molecular and cellular nature of organisms. At the center of the Biology undergraduate curriculum are the biochemical, genetic, and cell biology concepts and tools used to study the genes of prokaryotes and eukaryotes, to study the proteins and ribonucleic acids (RNA) encoded by these genes, and to study how the expression of these genes is regulated during the development and lifetimes of organisms. Molecular Biology represents a fusion of the four disciplines of biochemistry, biophysics, genetics, and cell biology. Modern biology requires a background in other disciplines such as chemistry, mathematics, physics, and computer sciences. Principles from these disciplines have to be merged to understand and apply new biotechnology and genetic engineering techniques. It is desirable for entering students to have a broad interest and background in the sciences.

Molecular Biology (BS)

A BS degree is offered in Molecular Biology. The BS degrees are intended as preparation for scientific careers in biology or careers in the health professions. Biology offers a streamlined double major with Business Administration or Criminology. Fast Track BS / MS Biology and Molecular Biology degree programs are available.

Bachelor of Science in Molecular Biology

[Degree Requirements](#) (120 semester credit hours)¹

[View an Example of Degree Requirements by Semester](#)

Faculty

FACG> nsm-molecular-biology-bs

Professors: Rockford K. Draper, Juan E. González, Lawrence J. Reitzer, Stephen Spiro, Li Zhang, Michael Qiwei Zhang

Associate Professors: Jeff L. DeJong, Nikki Delk, Heng Du, Tae Hoon Kim, Faruck Morcos, Kelli Palmer, Duane D. Winkler, Zhenyu Xuan

Assistant Professors: Nicole De Nisco, Nicholas Dillon, Purna Joshi, Jyoti Misra, Darshan Sapkota

Professors Emeriti: Hans Bremer, Lee A. Bulla, Donald M. Gray

Associate Professors Emeriti: Gail A. M. Breen, Dennis L. Miller

Clinical Professor: David Murchison

Research Assistant Professors: Lan Guo, Li Liu

Professors of Instruction: Scott A. Rippel, Uma Srikanth

Associate Professors of Instruction: Mehmet Candas, Wen-Ju Lin, Elizabeth Pickett, Ilya Sapozhnikov

Assistant Professors of Instruction: Ida Klang, Meenakshi Maitra, Caitlin Maynard, Iti Mehta, Ramesh Padmanabhan, Jing Pan, Ruben D. Ramirez, Eva Sadat, Subha Sarcar, Michelle Wilson, Zhuoru Wu

Senior Lecturer: Wen-Ho Yu

I. Core Curriculum Requirements: 42 semester credit hours²

Communication: 6 semester credit hours

[COMM 1311](#) Survey of Oral and Technology-based Communication

[RHET 1302](#) Rhetoric

Or select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2417](#) Calculus I^{3, 4, 5}

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[CHEM 1311](#) General Chemistry I³

or [CHEM 1315](#) Honors Freshman Chemistry I³

[CHEM 1312](#) General Chemistry II³

or [CHEM 1316](#) Honors Freshman Chemistry II³

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 3 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

[GOVT 2305](#) American National Government

[GOVT 2306](#) State and Local Government

Or select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[BIOL 2311](#) Introduction to Modern Biology I ₃

[MATH 2419](#) Calculus II _{3, 4, 5}

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 69-70 semester credit hours

Major Preparatory Courses: 24-25 semester credit hours beyond Core Curriculum

[CHEM 1111](#) General Chemistry Laboratory I

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 1311](#) General Chemistry I ₃

or [CHEM 1315](#) Honors Freshman Chemistry I ₃

[CHEM 1312](#) General Chemistry II₃

or [CHEM 1316](#) Honors Freshman Chemistry II₃

[CHEM 2323](#) Introductory Organic Chemistry I₆

or [CHEM 2327](#) Honors Organic Chemistry I₆

[CHEM 2325](#) Introductory Organic Chemistry II₆

or [CHEM 2328](#) Honors Organic Chemistry II₆

[CHEM 2233](#) Introductory Organic Chemistry Laboratory₆

or [CHEM 2237](#) Honors Organic Chemistry Laboratory₆

[MATH 2417](#) Calculus I_{3, 4, 5}

[MATH 2419](#) Calculus II_{3, 4, 5}

[MATH 2418](#) Linear Algebra

[PHYS 2325](#) Mechanics and [PHYS 2125](#) Physics Laboratory I

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat₇

[PHYS 2326](#) Electromagnetism and Waves

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves

[PHYS 2126](#) Physics Laboratory II

Major Core Courses: 33 semester credit hours beyond Core Curriculum

[BIOL 2111](#) Introduction to Modern Biology Workshop I₆

[BIOL 2112](#) Introduction to Modern Biology Workshop II₆

[BIOL 2281](#) Introductory Biology Laboratory₆

[BIOL 2311](#) Introduction to Modern Biology I_{3, 6}

[BIOL 2312](#) Introduction to Modern Biology II₆

[BIOL 3101](#) Classical and Molecular Genetics Workshop

[BIOL 3102](#) Eukaryotic Molecular and Cell Biology Workshop

[BIOL 3161](#) Biochemistry Workshop I

[BIOL 3162](#) Biochemistry Workshop II

[BIOL 3301](#) Classical and Molecular Genetics

[BIOL 3302](#) Eukaryotic Molecular and Cell Biology

[BIOL 3361](#) Biochemistry I

[BIOL 3362](#) Biochemistry II

or [BIOL 3335](#) Microbial Physiology

[BIOL 3380](#) Biochemistry Laboratory

[BIOL 4380](#) Cell and Molecular Biology Laboratory⁸

or [BIOL 3V96](#) Undergraduate Research in Molecular and Cell Biology⁸ (3 semester credit hours)

or [BIOL 4399](#) Senior Honors Research for Thesis in Molecular and Cell Biology⁸

or [BIOL 4391](#) Senior Research in Molecular and Cell Biology⁸

[BIOL 4461](#) Biophysical Chemistry

Major Related Courses: 12 semester credit hours⁹

12 semester credit hours upper-division approved molecular biology-related BIOL or CHEM electives

III. Elective Requirements: 8-9 semester credit hours

Free Electives: 8-9 semester credit hours

All students must complete at least 51 semester credit hours of upper-division courses to graduate.

Fast Track Baccalaureate/Master's Degrees

UT Dallas undergraduate students with strong academic records, including at least 15 semester credit hours of upper-division Biology core courses, who intend to pursue graduate work in Biology at UT Dallas, may apply for the Fast Track which involves taking selected graduate courses as an upper-division student. After Fast Track admission to the graduate program, 15 semester credit hours of graduate courses with an earned grade of B or better can be used toward completion of the BS and to satisfy requirements for those courses at the graduate level. Graduate courses must be approved by the graduate advisor. This program provides an opportunity to obtain the BS degree in Biology after 120 semester credit hours of work and an MS degree in Molecular and Cell Biology after an additional 21 semester credit hours of graduate course and research work. Interested students should contact the Biology undergraduate advisor well in advance of the senior year to prepare a degree plan taking maximal advantage of this Fast Track program.

Degree Planning

Upper-division biology courses taken at other institutions may be included as part of the degree plan subject to the provisions of the section on Transfer Admissions.

Major-related courses may not include more than 9 semester credit hours (BS) or 6 semester credit hours (BA) of upper-division transfer credit and not more than 3 semester credit hours (Biology major) or 6 semester credit hours (Molecular Biology major) of individual instruction (e.g., [BIOL 3V90](#), [BIOL 3V91](#), [BIOL 3V96](#), [BIOL 4302](#), [BIOL 4390](#), [BIOL 4391](#), [BIOL 4399](#), or [BIOL 4V99](#)).

Students planning a career in a particular allied health profession should consult the school they expect to attend to apprise themselves of the course requirements for admission.

Admission standards for medical and dental schools are set by the individual professional school, whose specific requirements should be reviewed with the help of the UT Dallas Health Professions Advising Center (HPAC). Most professional schools prefer that admission applications be channeled through the HPAC.

1. Incoming freshmen must enroll and complete requirements of UNIV 1010 and the corresponding school-related freshman seminar course. Students, including transfer students, who complete their core curriculum at UT Dallas must take UNIV 2020.
2. Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.
3. A required Major course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.

4. Six semester credit hours of Calculus are counted under Mathematics Core and Component Area Option and 2 semester credit hours of Calculus are counted as Major Preparatory Courses.
5. Students may choose one of the following calculus sequences: (a) MATH 2413, MATH 2414, and MATH 2415; or (b) MATH 2417 and MATH 2419.
6. Indicates a prerequisite class to be completed before enrolling for upper-division classes.
7. Students who complete PHYS 2421 do not need to complete PHYS 2125.
8. These substitutes for BIOL 4380 require permission of the Biology Undergraduate Faculty Advisor to ensure equivalent training in recombinant DNA analysis.
9. Up to 6 semester credit hours of research may be used in fulfilling the major related course requirement.

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